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Deliverable 2: Implementation of Database System with MongoDB **50 Marks**

Study the scenario 1 and complete the question(s) that follow:

MongoDB Implementation: Modise Cultural Facility Management System

Implement the Modise Cultural Facility scenario using MongoDB. Design an appropriate data model that captures the hierarchical relationships among provinces, municipalities, facilities, rooms, and activities. You may choose between embedding related documents or referencing them across collections. Ensure that your implementation supports efficient querying for activities by date, facility details, and their associated municipalities and provinces. Provide sample documents and queries that demonstrate the functionality of your data model. The following sample data may be considered in the database implementation. The following sample data may be considered for implementation of the database system.

Province	
provinceCode	name
WC	Western Cape
KZN	KwaZulu-Natal

Municipality			
munCode	name	avgPopulation	provinceCode
CPT	City of Cape Town	4005016	WC
ETH	eThekwini Metropolitan Municipality	3702231	KZN

Facility				
facilityId	name	address	capacity	munCode

Arts	Artscape Theatre Centre	D.F. Malan St, Foreshore, Cape Town, 8001	1500	CPT
ICC	Durban ICC	45 Bram Fischer Rd, Durban Central, Durban, 4001	10000	ETH

Room		
roomNo	description	facilityId
1	Hall 1	Arts
2	Hall 2	ICC

Activity	
activityRef	name
MUS01	Music Concert
EXH01	Art Exhibition

USES		
facilityId	activityRef	Date
Arts	MUS01	2025-05-20
ICC	EXH01	2025-06-15

Sample data for consideration.

Source: Obaro Adewale (2025)

Note:

- All questions must be answered using queries only. All queries must be typed in formatted text. As such, screenshot(s)/images of queries is completely unacceptable and would attract zero.
- All queries in this section must be written in MongoDB DBMS platform only.
- Queries written using another database IDE apart from MongoDB platform would automatically attract zero.

Questions for Consideration Under Deliverable 2

Use the above scenario to answer the following questions:

2.1 Create a database called **OLMS** using **query**.

(1 Mark)

2.2 Create the collections identified / derived from the ERD in the given scenario.

(5 Marks)

2.3 Store at least **two (2)** records into each collection in Question 2.2.

(18 Marks)

2.4 Write a query to retrieve and display the name of a facility such as “Arts” and all the related rooms associated with the facility.

2.5 Write a query to retrieve and display a list of municipalities with an average population of 3,500,000 (3.5 million) or more.

2.6 Retrieve and display the date on which a facility is used, the details of the facility including facility identity, name, capacity and address.

2.7 Write a query to retrieve and display a list of activities including activity reference codes, activity name along with the date on which each activity takes place as well as

detailed information regarding the facility such as the facility identity, facility name, capacity and address of each facility. In addition, the municipalities and province in which the facility are situated should be displayed with other requested information. This task requires obtaining information from multiple collections.

Question 2

```
show databases
```

```
admin   40.00 KiB
```

```
config 108.00 KiB
```

```
local   40.00 KiB
```

2.1 Create database in query

```
use OLMS
```

```
switched to db OLMS
```

```
show dbs
```

```
admin   40.00 KiB
```

```
config 108.00 KiB
```

```
local   40.00 KiB
```

2.2 Create Collections:

```
db.createCollection("Province")
```

```
{ ok: 1 }
```

```
db.createCollection("Municipality")
```

```
{ ok: 1 }
```

```
db.createCollection("Facility")
```

```
{ ok: 1 }
```

```
db.createCollection("Room")
```

```
{ ok: 1 }
```

```
db.createCollection("Activity")
```

```
{ ok: 1 }
```

```
db.createCollection("Uses")
```

```
{ ok: 1 }
```

2.3 Storing records in each collection:

```
db.Province.insertMany([{{provinceCode: "WC", name: "Western Cape"},{provinceCode: "KZN", name: "KwaZulu-Natal"}}])
```

```
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('684f05e19c595ce9530dbfec'),  
    '1': ObjectId('684f05e19c595ce9530dbfed')  
  }  
}
```

```
db.Municipality.insertMany([{{munCode: "CPT", name: "City of Cape Town", avgPopulation: 4005016, provinceCode: "WC"},{munCode: "ETH", name: "eThekweni Metropolitan Municipality", avgPopulation: 3702231, provinceCode: "KZN"}}])
```

```
{  
  acknowledged: true,  
  insertedIds: {  
    '0': ObjectId('684f09019c595ce9530dbfee'),  
    '1': ObjectId('684f09019c595ce9530dbfef')  
  }  
}
```

```
db.Facility.insertMany([{{facilityId: "Arts", name: "Artcape Theatre Centre", address: "D.F. Malan St, Foreshore, Cape Town, 8001", capacity: 1500, munCode: "CPT"},{facilityId: "ICC", name: "Durban ICC", address: "45 Bram Fischer Rd, Durban Central, Durban, 4001", capacity: 10000, munCode: "ETH"}}])
```

```
{
```

```

acknowledged: true,
insertedIds: {
  '0': ObjectId('684f0b009c595ce9530dbff0'),
  '1': ObjectId('684f0b009c595ce9530dbff1')
}
}

db.Room.insertMany([{roomNo: 1, description: "Hall 1", facilityId: "Arts"},{roomNo: 2,
description: "Hall 2", facilityId: "ICC"}])

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('684f0d919c595ce9530dbff2'),
    '1': ObjectId('684f0d919c595ce9530dbff3')
  }
}

db.Activity.insertMany([{activityRef: "MUS01", name: "Music Concert"},{activityRef:
"EXH01", name: "Art Exhibition"}])

{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('684f0e3d9c595ce9530dbff4'),
    '1': ObjectId('684f0e3d9c595ce9530dbff5')
  }
}

db.Uses.insertMany([{facilityId: "Arts", activityRef: "MUS01", Date: ISODate("2025-05-
20")},{facilityId: "ICC", activityRef: "EXH01", Date: ISODate("2025-06-15")}]])

{
  acknowledged: true,

```

```
insertedIds: {  
  '0': ObjectId('684f0f4f9c595ce9530dbff6'),  
  '1': ObjectId('684f0f4f9c595ce9530dbff7')  
}  
}
```

OLMS

2.4. Query to display names of facility and their related rooms associated with the facility:

```
db.Room.find()
```

```
{  
  _id: ObjectId('684f0d919c595ce9530dbff2'),  
  roomNo: 1,  
  description: 'Hall 1',  
  facilityId: 'Arts'  
}
```

```
{  
  _id: ObjectId('684f0d919c595ce9530dbff3'),  
  roomNo: 2,  
  description: 'Hall 2',  
  facilityId: 'ICC'  
}
```

```
db.Room.updateOne({roomNo: 1, facilityId: "Arts"}, {$set: {description: "Main Theater"}})
```

```
{  
  acknowledged: true,  
  insertedId: null,  
  matchedCount: 1,
```



```

    modifiedCount: 1,
    upsertedCount: 0
  }
  db.Room.insertOne({roomNo: 2, description: "Opera House", facilityId: "Arts"})

  {
    acknowledged: true,
    insertedId: ObjectId('6851d7b2fa5728f3369e6661')
  }

  db.Facility.aggregate([{$match: {facilityId: "Arts"}},{$lookup: {from: "Room", localField:
"facilityId", foreignField: "facilityId", as: "rooms"}}, {$project: {_id: 1, facilityId: 1, name: 1,
capacity: 1, address: 1, munCode: 1, rooms: {$map: {input: "$rooms", as: "room", in:
{roomNo: "$$room.roomNo", description: "$$room.description"}}}}})

  {
    _id: ObjectId('684f0b009c595ce9530dbff0'),
    facilityId: 'Arts',
    name: 'Artcape Theatre Centre',
    address: 'D.F. Malan St, Foreshore, Cape Town, 8001',
    capacity: 1500,
    munCode: 'CPT',
    rooms: [
      {
        roomNo: 1,
        description: 'Main Theater'
      },
      {
        roomNo: 2,
        description: 'Opera House'
      }
    ]
  }

```

```

    }
  ]
}

```

OLMS

2.5 Query to display list of municipalities with average population of greater and equal to 3.5 million:

```
db.Municipality.aggregate([{$match: {avgPopulation: {$gte: 3500000}}},{$project: {_id: "$munCode", name: 1, avgPopulation: 1, provinceCode: 1}}])
```

```

{
  name: 'City of Cape Town',
  avgPopulation: 4005016,
  provinceCode: 'WC',
  _id: 'CPT'
}
{
  name: 'eThekweni Metropolitan Municipality',
  avgPopulation: 3702231,
  provinceCode: 'KZN',
  _id: 'ETH'
}

```

OLMS

2.6 Query to display the dates the facility will be used:

```
db.Uses.aggregate([{$lookup: {from: "Facility", localField: "facilityId", foreignField: "facilityId", as: "facilityDetails"}},{$unwind: "$facilityDetails"},{$project: {_id: 0, useDate: "$Date", facilityId: "$facilityDetails.facilityId", facilityName: "$facilityDetails.name", capacity: "$facilityDetails.capacity", address: "$facilityDetails.address"}}])
```

```

{

```

```

    useDate: 2025-05-20T00:00:00.000Z,
    facilityId: 'Arts',
    facilityName: 'Artcape Theatre Centre',
    capacity: 1500,
    address: 'D.F. Malan St, Foreshore, Cape Town, 8001'
  }
  {
    useDate: 2025-06-15T00:00:00.000Z,
    facilityId: 'ICC',
    facilityName: 'Durban ICC',
    capacity: 10000,
    address: '45 Bram Fischer Rd, Durban Central, Durban, 4001'
  }
}
OLMS

```

2.7 Query to display list of activities :

```

db.Uses.aggregate([{$lookup: {from: "Activity", localField: "activityRef", foreignField:
"activityRef", as: "activity"}},{$unwind: "$activity"},{$lookup: {from: "Facility", localField:
"facilityId", foreignField: "facilityId", as: "facility"}},{$unwind: "$facility"},{$lookup: {from:
"Municipality", localField: "facility.munCode", foreignField: "munCode", as:
"municipality"}},{$unwind: "$municipality"},{$lookup: {from: "Province", localField:
"municipality.provinceCode", foreignField: "provinceCode", as: "province"}},{$unwind:
"$province"},{$project: { _id: 0, useDate: "$Date", activityRef: 1, activityName:
"$activity.name", facilityId: "$facility.facilityId", facilityName: "facility.name", capacity:
"$facility.capacity", address: "$facility.address", municipality: "$municipality.name",
province: "$province.name"}}})
{
  activityRef: 'MUS01',
  useDate: 2025-05-20T00:00:00.000Z,
  activityName: 'Music Concert',

```

```
facilityId: 'Arts',
facilityName: 'facility.name',
capacity: 1500,
address: 'D.F. Malan St, Foreshore, Cape Town, 8001',
municipality: 'City of Cape Town',
province: 'Western Cape'
}
{
activityRef: 'EXH01',
useDate: 2025-06-15T00:00:00.000Z,
activityName: 'Art Exhibition',
facilityId: 'ICC',
facilityName: 'facility.name',
capacity: 10000,
address: '45 Bram Fischer Rd, Durban Central, Durban, 4001',
municipality: 'eThekweni Metropolitan Municipality',
province: 'KwaZulu-Natal'
}
OLMS
```

Deliverable 3: Implementation of Database System with Neo4j DBMS

50 Marks

Study the scenario 1 and complete the question(s) that follow:

Neo4j Implementation: Modise Cultural Facility Management System

Implement the Modise Cultural Facility scenario using Neo4j dbms that accurately represents entities such as provinces, municipalities, facilities, rooms, and activities as nodes, and their interconnections as relationships. Ensure that your model leverages Neo4j's strengths by using specific relationship types and appropriate directionality to reflect the nature of the connections. Incorporate relevant properties for both nodes and relationships to capture essential details, such as activity dates or facility capacities. Provide sample Cypher queries that demonstrate the retrieval of complex information, such as activities hosted by facilities within specific municipalities and provinces on given dates. The following sample data may be considered for implementation of the database system.

Province	
provinceCode	name
GP	Gauteng
WC	Western Cape

Municipality			
munCode	name	averagePopulation	provinceCode
JHB	City of Johannesburg	4949347	GP
CPT	City of Cape Town	4005016	WC

Facility				
facilityId	name	address	capacity	munCode
FAC001	Modise Cultural Facility	123 Cultural St	500	JHB

FAC002	Ubuntu Arts Centre	456 Ubuntu Ave	300	CPT
--------	--------------------	----------------	-----	-----

Room		
roomNo	description	facilityId
R001	Theatre	FAC001
R002	Cinema	FAC001
R003	Exhibition Hall	FAC002
R004	Workshop Room	FAC002

Activity	
activityRef	name
ACT001	IsiZulu Dance Competition
ACT002	Art Exhibition
ACT003	Music Concert
ACT004	Theatre Performance

USES		
facilityId	activityRef	Date
FAC001	ACT001	2025-05-20
FAC001	ACT002	2025-06-20
FAC002	ACT003	2025-10-20
FAC002	ACT004	2025-12-12

Sample data for consideration.

Source: Obaro Adewale (2025)

Note:

- All questions must be answered using queries only. All queries must be typed in formatted text. As such, screenshot(s)/images of queries is completely unacceptable and would attract zero.
- All queries in this section must be written in Neo4j DBMS platform only.
- Queries written using another database IDE apart from Neo4j platform would automatically attract zero.

Questions for Consideration Under Deliverable 3

Use the information provided to answer the following questions:

3.1 Create a database called ONLMS **using query**.

(2 Marks)

3.2 Using the provided sample data or sample data of choice, store at least two (2) data nodes and establish the relationships that exist between the collections (tables):

(a) Data nodes of province and municipality as well as related relationship.

(b) Data nodes of municipality and facility as well as related relationship.

(c) Data nodes of facility and room as well as associated relationship.

(d) Data nodes of facility and activity room as well as associated relationship.

3.3 Using query, display all the created collections (nodes) and their relationships in a graph data model as shown in the figure below. The result output of the query for **all nodes and relationships in graphical** may be displayed in the format below.

3.4 Create a script to display the total capacity of facilities within each municipality.

3.5 Retrieve and display a list of all facilities along with their municipalities and provinces. The result of the query/script obtained from three (3) nodes (collections) may be displayed in any of the format below.

3.6 Write a query to determine and display all activities along with the number of days until they occur from today.

3.7 Write a query to retrieve and display a list of municipalities with an average population of 4,000,000 (4 million) or more.

Question 3

3.1 **create database** ONLMS

3.2 a) Here we have created the nodes 'Province' and 'Municipality'. The reason why 'provinceCode' was not included in the 'Municipality' node is because we want to create a relationship between 'Province' and 'Municipality' and we want to avoid duplicate data in the database. The relationship ':HAS_MUNICIPALITY' has been form/created.

:use ONLMS

```
CREATE (:Province {provinceCode: 'GP', name: 'Gauteng'}), (:Province {provinceCode: 'WC', name: 'Western Cape'}),  
(:Municipality {munCode: 'JHB', name: 'City of Johannesburg', averagePopulation: 4949347}), (:Municipality {munCode: 'CPT', name: 'City of Cape Town', averagePopulation: 4005016});
```

```
MATCH (pro1:Province {name: 'Gauteng'}), (mun1:Municipality {name: 'City of Johannesburg'})
```

```
MERGE (pro1)-[:HAS_MUNICIPALITY]->(mun1);
```

```
MATCH (pro2:Province {name: 'Western Cape'}), (mun2:Municipality {name: 'City of Cape Town'})
```

```
MERGE (pro2)-[:HAS_MUNICIPALITY]->(mun2);
```

b) First created facility node and then the relationship between 'Municipality' and 'Facility'


```
create (:Facility {FacilityId: 'FAC001', name: 'Modise Cultural Facility', address: '123 Cultural St', capacity: 500 }), (:Facility {facilityId: 'FAC002', name: 'Ubuntu Arts Centre', address: '456 Ubuntu Ave', capacity: 300});
```

```
MATCH (mun1:Municipality {name: 'City of Johannesburg'}), (fac1:Facility {name: 'Modise Cultural Facility'})
```

```
MERGE (mun1)-[:HAS_FACILITY]->(fac1);
```

```
MATCH (mun2:Municipality {name: 'City of Cape Town'}), (fac2:Facility {name: 'Ubuntu Arts Centre'})
```

```
MERGE (mun2)-[:HAS_FACILITY]->(fac2);
```

To view:

```
MATCH (mun:Municipality)-[r:HAS_FACILITY]->(fac:Facility)
```

```
RETURN mun, r, fac
```

c) Created the room node and then the relationship between 'Facility' and 'Room'

```
create (:Room {roomNo: 'R001', description: 'Theatre'}), (:Room {roomNo: 'R002', description: 'Cinema'}), (:Room {roomNo: 'R003', description: 'Exhibition Hall'}), (:Room {roomNo: 'R004', description: 'Workshop Room'});
```

```
MATCH (fac1:Facility {name: 'Modise Cultural Facility'}), (room1:Room {description: 'Theatre'})
```

```
MERGE (fac1)-[:HAS_ROOM]->(room1);
```

```
MATCH (fac1:Facility {name: 'Modise Cultural Facility'}), (room2:Room {description: 'Cinema'})
```

```
MERGE (fac1)-[:HAS_ROOM]->(room2);
```

```
MATCH (fac2:Facility {name: 'Ubuntu Arts Centre'}), (room3:Room {description: 'Exhibition Hall'})
```

```
MERGE (fac2)-[:HAS_ROOM]->(room3);
```

```
MATCH (fac2:Facility {name: 'Ubuntu Arts Centre'}), (room4:Room {description: 'Workshop Room'})
```

```
MERGE (fac2)-[:HAS_ROOM]->(room4);
```

To view:

```
MATCH p=()-[r:HAS_ROOM]->() RETURN p LIMIT 25
```

d) Created the 'Activity' and formed a relationship between 'Facility' and 'Activity'

```
create (:Activity {activityRef: 'ACT001', name: 'isiZulu Dance Competition'}), (:Activity {activityRef: 'ACT002', name: 'Art Exhibition'}), (:Activity {activityRef: 'ACT003', name: 'Music Concert'}), (:Activity {activityRef: 'ACT004', name: 'Theatre Performance'});
```

```
MATCH (fac1:Facility {name: 'Modise Cultural Facility'}), (act1:Activity {name: 'isiZulu Dance Competition'})
```

```
MERGE (fac1)-[:HOSTS_ON_2025_05_20]->(act1);
```

```
MATCH (fac1:Facility {name: 'Modise Cultural Facility'}), (act2:Activity {name: 'Art Exhibition'})
```

```

MERGE (fac1)-[:HOSTS_ON_2025_06_20]->(act2);
MATCH (fac2:Facility {name: 'Ubuntu Arts Centre'}), (act3:Activity {name: 'Music Concert'})
MERGE (fac2)-[:HOSTS_ON_2025_10_20]->(act3);
MATCH (fac2:Facility {name: 'Ubuntu Arts Centre'}), (act4:Activity {name: 'Theatre Performance'})
MERGE (fac2)-[:HOSTS_ON_2025_12_12]->(act4);

```

To view :

```
MATCH p=()-[r:HOSTS_ON_2025_05_20]->() RETURN p LIMIT 25
```

```
MATCH p=()-[r:HOSTS_ON_2025_06_20]->() RETURN p LIMIT 25
```

```
MATCH p=()-[r:HOSTS_ON_2025_10_20]->() RETURN p LIMIT 25
```

```
MATCH p=()-[r:HOSTS_ON_2025_12_12]->() RETURN p LIMIT 25
```

3.3 The result output of the query for all nodes and relationships :

```

MATCH (n)-[r]->(m)
RETURN n, r, m

```

3.4 query to display the total capacity of facilities:

```

MATCH (mun:Municipality)-[:HAS_FACILITY]->(fac:Facility)
RETURN mun.name AS MunicipalityName, SUM(fac.capacity) AS TotalCapacity
ORDER BY TotalCapacity DESC;

```

3.5 Query to display all facilities with their municipalities and provinces:

```

MATCH (pro:Province)-[:HAS_MUNICIPALITY]->(mun:Municipality)-[:HAS_FACILITY]->(fac:Facility)
RETURN fac.name AS FacilityName, mun.name AS MunicipalityName, pro.name AS ProvinceName
ORDER BY ProvinceName, MunicipalityName, FacilityName;

```

3.6 Query to display all activities and days until the activity occur from **TODAY**:

```

WITH date() AS today
MATCH (fac:Facility)-[r]->(act:Activity)
WITH fac.name AS FacilityName, act.name AS ActivityName, split(type(r), "_ON_")[1] AS rawDate, today
WITH FacilityName, ActivityName, toInteger(split(rawDate, "_")[0]) AS year, toInteger(split(rawDate, "_")[1]) AS month, toInteger(split(rawDate, "_")[2]) AS day, today
WITH FacilityName, ActivityName, date({year: year, month: month, day: day}) AS ActivityDate, today
RETURN ActivityName, FacilityName, ActivityDate, duration.between(today, ActivityDate).days AS DaysUntilActivity
ORDER BY DaysUntilActivity

```

3.7 Query to display a list of municipalities with average Population of greater and equal to 4 million:

```
MATCH (mun:Municipality) WHERE mun.averagePopulation >= 4000000
RETURN mun.munCode AS MunicipalityCode, mun.name AS MunicipalityName, mun.
averagePopulation AS AveragePopulation
ORDER BY AveragePopulation DESC
```